Part Number	74-8812-DS	**Document No.**	WI-0744
Part Name	Drive Shaft Sub-Assembly	**Revision**	A
Operation	Op 50: Bearing Press, Grease Fill, and Balance	**Effective Date**	2026-10-05
Work Center	Press Station 9 / Balancer 2	**Takt / Cycle Time**	8.5 min
Prepared By	Manufacturing Engineering	**Approved By**	Quality Manager

1. Purpose and Scope

This work instruction defines bearing installation, snap ring retention, grease fill, and dynamic balancing of drive shaft sub-assembly 74-8812-DS prior to final assembly.

2. Safety Requirements and PPE

Requirement	Detail
PPE	Safety glasses, nitrile gloves for grease handling, steel-toe footwear.
Hydraulic press	Light curtain must be active. Never reach into the press area while the ram is in motion.
Rotating equipment	The balancer guard must be closed before the spin cycle will start. Keep loose clothing and lanyards secured.
Grease	Wipe spills immediately. Grease on the floor is a slip hazard.

3. Tools, Equipment, and Materials

Item	Description	Spec / Calibration
Hydraulic press	Station 9 press with force monitor	Force curve monitor active
Press tooling	Adapter set PT-7488 (bearing OD contact only)	Correct adapter per photo card
Snap ring pliers	External, 90 degree tip	Tips not worn
Grease dispenser	Metered dispenser, Grease spec LG-2EP	Shot size verified at shift start
Dynamic balancer	Balancer 2 with 74-8812 drive fixture	Daily master rotor check
Dial indicator	0.01 mm, with magnetic base	Calibrated

4. Components / Materials Required

Component	Part / Spec No.	Qty
Shaft, machined	74-8810 (Op 40 complete)	1
Ball bearing, sealed	BR-6205-2RS	1
Snap ring, external 25 mm	SR-2500-E	1
Grease, lithium EP2	LG-2EP	12 g
Balance weights, clip-on	BW-7400 series	As req'd

5. Work Steps

Step	Instruction	Key Point / Specification
1	Pull one shaft from the inbound rack and wipe the bearing journal with a clean towel. Inspect the journal for nicks or scoring.	Journal clean, no damage
2	Verify the bearing box lot number and that the seals face the correct way per the photo card (red seal toward the shaft shoulder).	Red seal faces shoulder
3	Load the shaft vertically into the press fixture, journal up.	Shaft seated on datum ring
4	Place the bearing square on the journal and position press adapter PT-7488 on the bearing outer race only.	Adapter contacts OD race only
5	Cycle the press. Watch the force monitor during the stroke.	Force 8 to 12 kN within window
6	If the force curve alarms (under or over window), stop and tag the assembly. Do not re-press.	No re-press allowed
7	Verify the bearing is seated against the shoulder using a 0.05 mm feeler gage. The feeler must not enter at any point.	0.05 mm feeler: no-go
8	Install the external snap ring into the groove with snap ring pliers. Rotate the ring to confirm full seating in the groove.	Ring fully in groove, rotates freely
9	Dispense grease into the coupling pocket using the metered dispenser, one shot.	12 +/- 1 g per shot
10	Load the assembly into Balancer 2, close the guard, and run the balance cycle.	Guard latched, auto cycle
11	If imbalance exceeds the limit, install clip-on weights at the plane and angle shown on the balancer screen, then re-run.	Residual imbalance 15 g-mm max
12	Maximum two correction passes. If still out of limit, tag as nonconforming.	Max 2 correction passes
13	With the assembly still in the fixture, check runout at the mid-span with the dial indicator while rotating by hand.	0.08 mm TIR max
14	Apply the green QC dot to the shaft end face and place the assembly horizontally in the padded outbound cart.	Green dot, padded cart

6. Quality Checks

Characteristic	Specification	Method	Frequency
Press force	8 to 12 kN	Force monitor trace	Every part
Bearing seating	0.05 mm feeler no-go	Feeler gage	Every part
Residual imbalance	15 g-mm max	Balancer readout	Every part
Runout, mid-span	0.08 mm TIR max	Dial indicator	Every part
Grease shot audit	12 +/- 1 g	Scale check	1 per 25 pcs

7. Nonconforming Material

If any check fails or a defect is observed, STOP the operation. Tag the part with a red NONCONFORMING tag, record the part number, lot, and defect on the tag, and move the part to the quarantine rack. Notify the line lead and Quality immediately. Do not rework or scrap without an approved disposition.

8. Revision History

Rev	Date	Description	Approved
A	2026-10-05	Initial release	QM

Operator Signature: _____ Date: _____

Supervisor Verification: _____ Date: _____